Quantitative Methods I

Week 4 - Finding the Derivative

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Motivation

The difference quotient approach

- leads to the derivative, but
- is a lot of work

Easier rules

- such as for power functions are appreciated
- and more general rules do exist

Arithmetic Rules for the Derivative

THE PRODUCT RULE

If the derivatives f' and g' exist, then

$$[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

THE QUOTIENT RULE

If the derivatives f' and g' exist, then

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Applying the Product Rule

THE PRODUCT RULE

If the derivatives f' and g' exist, then

$$[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

Consider

$$h(x) = (x^3 - x)(5x^4 + x^2)$$

Then, $h(x) = f(x) \cdot g(x)$, for

$$f(x) = (x^3 - x) \qquad \text{and} \qquad g(x) = (5x^4 + x^2)$$

Since,

$$f'(x) = 3x^2 - 1 \qquad \text{and} \qquad g'(x) = (20x^3 + 2x)$$

the product rule implies that

$$h'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x) = (3x^2 - 1)(5x^4 + x^2) + (x^3 - x)(20x^3 + 2x)$$

Applying the Product Rule - cont'd

THE PRODUCT RULE

If the derivatives f' and g' exist, then

$$[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

- Demand d is a function of price p, i.e., $d(p)$
- And, revenue too, as it equals price times demand, i.e., $r(p) = p \cdot d(p)$
- Price elasticity of revenue is then:

$$\begin{aligned} El_r(p) &= r'(p) \cdot \frac{p}{r(p)} \\ &= [1 \cdot d(p) + p \cdot d'(p)] \cdot \frac{p}{p \cdot d(p)} \\ &= 1 + d'(p) \cdot \frac{p}{d(p)} = 1 + El_d(p) \end{aligned}$$

Applying the Quotient Rule

THE QUOTIENT RULE

If the derivatives f' and g' exist, then

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)\cdot g(x) - f(x)\cdot g'(x)}{(g(x))^2}$$

Consider

$$h(x) = \frac{3x-5}{x-2}$$

Then, $h(x) = \frac{f(x)}{g(x)}$, for

$$f(x) = 3x-5 \qquad \text{and} \qquad g(x) = x-2$$

Since,

$$f'(x) = 3 \qquad \text{and} \qquad g'(x) = 1$$

the quotient rule then implies that

$$h'(x) = \left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)\cdot g(x) - f(x)\cdot g'(x)}{(g(x))^2} = \frac{3\cdot(x-2)-(3x-5)\cdot 1}{(x-2)^2} = \frac{-1}{(x-2)^2}$$

The Quotient Rule - Practical Application(s)

THE QUOTIENT RULE

If the derivatives f' and g' exist, then

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

A company's total cost to produce q units is given by a function $C(q)$. The average cost per unit is therefore a quotient: $AC(q) = \frac{C(q)}{q}$

Question: At what production level is $AC(q)$ at its absolute minimum?

Example: If $C(q) = q^2 + 20q + 100$, then $AC(q) = \frac{q^2+20q+100}{q}$

Via the quotient rule, can find $AC'(q)$, solve for $AC'(q) = 0$, and identify the most cost-efficient level of production.

The Chain Rule

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

A function $h(x) = f(g(x))$ is a so-called composite function

- first, g is applied on x, then f is applied on $g(x)$
- here g is referred to as the inner and F as the outer function

Applying the Chain Rule

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Take, as an example, $h(x) = (2x+3)^4$

- first, we apply $y = g(x) = 2x + 3$
- then, we apply $z = f(y) = y^4$

Computing $h'(x) = [f(g(x))]'$ via the chain rule, we need

- the derivative of g, i.e., $g'(x) = 2$, and
- the derivative of f, $f'(y) = 4y^3$, but at $y = g(x)$, i.e., $f'(g(x)) = 4(2x+3)^3$

$$h'(x) = f'(g(x)) \cdot g'(x) = 4(2x+3)^3 \cdot 2 = 8(2x+3)^3$$

Applying the Chain Rule - cont'd

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Consider $h(x) = (1 - x^3)^5$

- define $g(x) = (1 - x^3)$ and $f(y) = y^5$ such that $h(x) = f(g(x))$
- then, $g'(x) = -3x^2$ and $f'(y) = 5y^4$ imply that $h'(x) = 5(1 - x^3)^4(-3x^2)$

Next, consider $k(x) = \sqrt{1 + x^2}$

- define $g(x) = 1 + x^2$ and $f(y) = \sqrt{y} = y^{\frac{1}{2}}$ such that $k(x) = f(g(x))$
- then, $g'(x) = 2x$ and $f'(y) = \frac{1}{2}y^{-\frac{1}{2}}$ imply that $k'(x) = \frac{1}{2}(1 + x^2)^{-\frac{1}{2}} \cdot 2x$

The Chain Rule - Practical Application(s)

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

A company's weekly revenue, R, depends on the number of sold units, q. But sales depend on the weekly advertising budget, a.

- Revenue is a function of sales: $R(q)$
- Sales are a function of advertising: $q(a)$

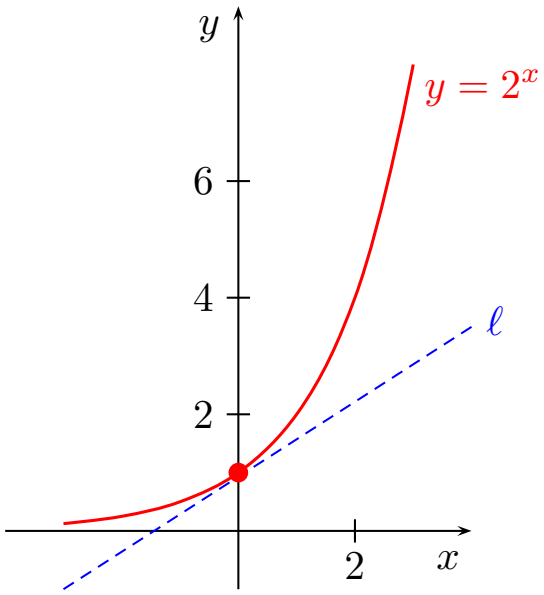
This creates a composite function: R(q(a)).

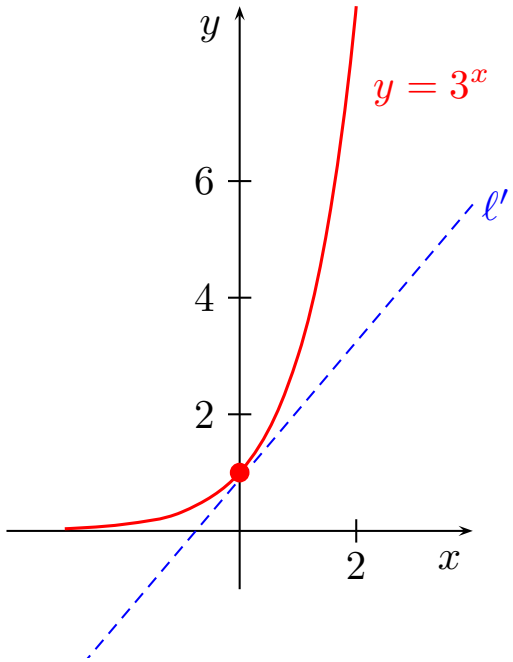
Question: If we increase a by 1 Euro, how does R change?

We need to find $\frac{dR}{da}$. By the chain rule, this is:

$$\frac{dR}{da} = \frac{dR}{dq} \cdot \frac{dq}{da}$$

Derivative of Exponential Functions





While the slope of ℓ is smaller than 1, the slope of ℓ' is larger than 1. As such, there exists an exponential function $y = f(x) = a^x$ where the tangent at the point $(x, y) = (0, 1)$ has slope 1. Indeed, this function is

$$[e^x]' = e^x$$

where $e \approx 2.718$ is the Euler number.

Derivative of Exponential Functions - cont'd

THE DERIVATIVE OF EXPONENTIAL FUNCTIONS

For $a > 0$,

$$[a^x]' = a^x \ln a$$

To see this, recall that, for $c > 0$,

- the exponential equation $b^x = c$ is solved by $x = {}^a\log c$
- as such, the equation $e^x = c$ is solved by $x = {}^e\log c = \ln c$
  - where ln is the "natural logarithm"
  - such that both $e^{\ln c} = c$ and $\ln(e^c) = c$
- This implies that $f(x) = a^x = [e^{\ln a}]^x = e^{x \ln a}$ and, thus,

$$f'(x) = (e^{x \ln a}) \cdot \ln a = a^x \ln a$$

Derivative of Logarithmic Functions

THE DERIVATIVE OF LOGARITHMIC FUNCTIONS

For $a > 0$ and $x > 0$,

$$[{}^a\log x]' = \frac{1}{x \ln a}$$

To see why, define $h(x) = f(g(x))$ where $y = g(x) = {}^a\log x$ and $f(y) = a^y$. Applying the Chain Rule, we have

$$h'(x) = [f(g(x))]' = \underbrace{a^{g(x)} \ln a}_{f'(g(x))} \cdot g'(x) = x \ln a \cdot g'(x)$$

where we have used that, by definition, it holds

$$a^{g(x)} = a^{{}^a\log x} = x$$

But, $h(x) = a^{g(x)}$. As such, $h'(x) = 1$ and, thus,

$$x \ln a \cdot g'(x) = 1 \qquad \text{or} \qquad g'(x) = \frac{1}{x \ln a}$$

Overview of Derivatives

$$
\begin{aligned}
f(x) &= x^r & \Rightarrow \quad & f'(x) = rx^{r-1} \\
f(x) &= e^x & \Rightarrow \quad & f'(x) = e^x \\
f(x) &= e^{g(x)} & \Rightarrow \quad & f'(x) = g'(x) \cdot e^{g(x)} \\
f(x) &= a^x & \Rightarrow \quad & f'(x) = a^x \ln a \\
f(x) &= \ln x & \Rightarrow \quad & f'(x) = \frac{1}{x} \\
f(x) &= \ln(g(x)) & \Rightarrow \quad & f'(x) = \frac{g'(x)}{g(x)} \\
f(x) &= {}^a\log x = \frac{\ln x}{\ln a} & \Rightarrow \quad & f'(x) = \frac{1}{x \ln y}
\end{aligned}
$$